

An experiment study for unmanned aerial manipulator systems with $\mathcal{L}_1$ adaptive augmentation of geometric control

Kexin Cai , Hai Yu , Zhaopeng Zhang, Xiao Liang ^{*}, Yongchun Fang, Jianda Han

Institute of Robotics and Automatic Information System, College of Artificial Intelligence, Nankai University, Tianjin, China

Tianjin Key Laboratory of Intelligent Robotics, Nankai University, Tianjin, China

Engineering Research Center of Trusted Behavior Intelligence, Ministry of Education, Nankai University, Tianjin, China

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ABSTRACT

Unmanned aerial manipulators, integrating multi-rotor unmanned aerial vehicles (UAVs) and multi-link manipulators, are extending the capabilities of UAVs to interact with the environment and demonstrating vast potential for diverse applications. However, the dynamic coupling between the UAV and the manipulator, along with the unmodeled nonlinear disturbances exerting on the aerial platform, pose challenges for high-performance control of the aerial manipulator. To this end, this paper proposes an $\mathcal{L}_1$ adaptive augmentation control method for the aerial manipulator to compensate the disturbances brought by manipulator motion and unmodeled uncertainties. Specifically, a baseline geometric controller is constructed to directly compensate the measurable wrench disturbances exerted on the UAV by the manipulator motion, while an $\mathcal{L}_1$ adaptive controller is designed to compensate the residual unknown wrench disturbances. By Lyapunov techniques, it is proven that the error signals of the closed-loop system are uniformly ultimately bounded, which demonstrates the theoretical effectiveness of the proposed method. Finally, the feasibility and robustness of the proposed control method are validated through three groups of hardware experiments.

1. Introduction

Unmanned aerial vehicles, known for their excellent maneuverability and flexibility (Jeong et al., 2024; Song et al., 2024; Xie et al., 2024; Yang et al., 2023), are frequently employed in passive scenarios such as geospatial mapping and security surveillance. With the advancement of aerial robotics technology, a new type of aerial robot composed of multi-rotor UAV and multi-link manipulator, known as an unmanned aerial manipulator (UAM), has expanded the applications of UAVs to active physical interactions with the environment. In recent years, aerial manipulators have been deployed for complex active tasks involving physical interactions, such as object grasping (Chen, Wu et al., 2022; Kim et al., 2016) and transportation (Liu et al., 2024; Orsag et al., 2017), aerial surface spraying (Vempati et al., 2018), switch pressing (Meng et al., 2018), contact inspection (Hu et al., 2022; Liang et al., 2024), dynamic environment interaction (Meng et al., 2025), and peg-in-hole task (Wang et al., 2024).

Compared to the control of a single multi-rotor UAV, the aerial manipulator system exhibits more complex nonlinearities and stronger dynamic coupling. The movement of the manipulator introduces challenges in system control due to the shifts in the center of mass (CoM)

and the dynamic disturbances. The motion of the manipulator generates reaction forces and torques, where coupling effects may result in inverse or delayed responses within the UAM system. Furthermore, in certain tasks such as grasping or manipulation, the aerial manipulator system must adhere to specific motion constraints, potentially introducing nonholonomic constraints. Consequently, the mathematical model of this system is highly complex, characterized by hard nonlinearities such as non-minimum phase behavior and nonholonomic constraints. Moreover, unmodeled nonlinear disturbances from the UAV further complicate the precise control of the UAM system. Currently, from a control perspective, the reported modeling and control work on aerial manipulators can be broadly categorized into coupled and decoupled approaches (Ollero et al., 2021). Coupled methods treat the entire aerial manipulator system as a unique entity, with controller designs that consider the full dynamics of the combined system. Decoupled methods, on the other hand, consider the UAV and the manipulator as two independently controlled subsystems, designing separate controllers for each part.

For coupled methods, the design relies on the full dynamic model. In Kim et al. (2016), the kinematic and dynamic models of the combined system are derived by considering the dynamic characteristics,

^{*} Corresponding author at: Institute of Robotics and Automatic Information System, College of Artificial Intelligence, Nankai University, Tianjin, China.
E-mail address: liangx@nankai.edu.cn (X. Liang).

and a passive adaptive controller is designed for both position and velocity control. In response to the complexity of the nonlinear dynamics of the aerial manipulator system, Kim et al. (2016) mention that neglecting the Coriolis matrix during controller design can simplify calculations, as this term is small but computationally complex during hover flight. In Suarez et al. (2017), the redundant degrees of freedom (DoFs) in the dual-arm prototype are utilized to generate coordinated motions during contact-less phase, so as to offset reaction torques partially. The design of the controller for the combined system in Suarez et al. (2017) requires information on the velocity and acceleration of the manipulator joints, which are typically difficult to measure for ordinary actuators. In Lee et al. (2020), a robust control method based on nonlinear disturbance observer (DOB) is proposed for the combined system, which achieves successful demonstrations on pushing a rolling cart and a hinged door. If the nonlinear dynamics of the combined system can be accurately modeled, coupled control methods have great prospects in terms of control accuracy and robustness (Ruggiero et al., 2018). However, real-time computation of the dynamic model of a large-scale nonlinear system and implementation of controller based on the full dynamic model require high-performance computing platforms. Moreover, coupled control methods require direct control of manipulator's motor moment, which makes it difficult to apply this method in aerial manipulator using traditional servo actuators.

Decoupled methods, due to their independence from the need to know the nonlinear model of the combined system and the lower computational costs, are more readily implemented in practice. Early work often overlooks the dynamic coupling effects, namely the influence of the manipulator on UAV dynamics, which tends to result in relatively lower control precision (Luna et al., 2022). As a result, researchers investigate the coupling effects brought by the manipulator motion. In Kondak et al. (2013), wrench sensors are installed between the autonomous helicopter and the manipulator to measure the interaction wrench, which is used in the subsequent controller design. However, the application of these expensive and heavy sensors is limited when it comes to small-scale multi-rotors. By utilizing the inverse dynamics of the manipulator, Jimenez-Cano et al. (2016) establishes a system model that contains variable inertia parameters, and an integral backstepping controller is designed to compensate the changes of the system's CoM. The method proposed in Jimenez-Cano et al. (2016) for compensating changes in the CoM can only counteract the manipulator's gravity, leading to challenges in achieving satisfactory tracking performance during rapid manipulator movements. In Zhang et al. (2019), a dynamic model of the multi-rotor equipped with a manipulator is established, the coupling effect model includes first and second derivatives of the variable inertia parameters, implying its applicability during rapid movements of the manipulator. However, traditional servo actuators often cannot directly provide the required first-order and second-order derivatives of joint angles. In recent years, machine learning-based control methods for aerial manipulators have been studied in Cao et al. (2022) and Liu and Huang (2021). In Liu and Huang (2021), a reinforcement learning method is adopted to control the manipulator, minimizing its impact on multi-rotor dynamics while following the desired trajectory. A motion controller based on adaptive neural networks is proposed in Cao et al. (2022), where a radial basis function neural network (RBFNN) with adaptive weight estimation laws is utilized to compensate the dynamic coupling between the multi-rotor and manipulator in real-time. Although machine learning-based control schemes can improve control performance, there are significant overheads associated with the data collection and training processes before deploying machine learning tools in the loop.

Considering the existence of model uncertainty, which can significantly degrade the control precision of the aerial manipulator, and the dynamic coupling disturbances caused by the manipulator's motion that exacerbate system fluctuations, the precise control of aerial manipulator faces substantial challenges. In light of these considerations, this paper proposes an $\mathcal{L}_1$ adaptive augmentation control scheme for

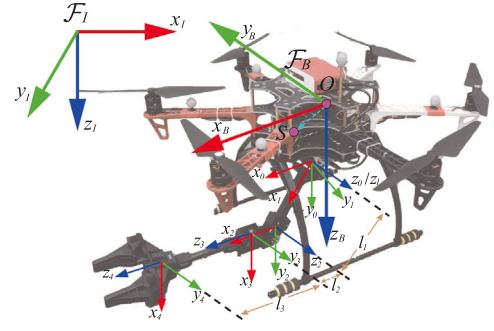


Fig. 1. Schematic diagram of the aerial manipulator system.

aerial manipulator to address disturbances caused by the manipulator's motion and unmodeled uncertainties. The motivation for utilizing $\mathcal{L}_1$ lies in its ability to compensate the uncertainties, which has been demonstrated in many works (Cheng et al., 2022; Hanover et al., 2021; Hovakimyan & Cao, 2010; Lakshmanan et al., 2020; Wu et al., 2023; Zuo & Mallikarjunan, 2016; Zuo & Ru, 2014). Specifically, for the aerial manipulator dynamics model containing unmeasurable velocity and acceleration signals of the system's CoM, the proposed scheme divides the disturbances into two parts through model transformation. The measurable part is compensated by utilizing feedforward technique, and an $\mathcal{L}_1$ adaptive controller is constructed to compensate the terms involving unmeasurable signals. Moreover, the proposed controller can also compensate the impact of unmodeled nonlinear disturbances without requiring explicit modeling of uncertainties. The stability of the proposed control method is analyzed via Lyapunov-based stability analysis, and hardware experiment results are provided to validate its performance. The main contributions of this paper are summarized as follows:

- (1) This paper constructs an $\mathcal{L}_1$ adaptive augmentation controller for the aerial manipulator system to compensate the uncertainties and dynamic coupling during manipulation tasks, and effectively resolves the issue in traditional control methods that neglect unmeasurable disturbances, thereby improving the system control precision when the manipulator moves.
- (2) The proposed $\mathcal{L}_1$ adaptive augmentation geometric control method is employed to robustify the aerial manipulator system, enabling tasks such as object grasping, transportation, and release, thereby expanding its application in a wider variety of scenarios.

The structure of this paper is organized as follows. Section 2 presents the dynamic model of the system. Section 3 elaborates on the development process of the controller. Stability analysis based on Lyapunov techniques is then conducted in Section 4. Section 5 involves three sets of hardware experimental tests to verify the performance of the controller. Finally, Section 6 concludes this paper.

2. Model transformation

As shown in Fig. 1, the aerial manipulator system consists of a hex-rotor UAV and a 3-DOF manipulator. The inertial frame is denoted as F_I , and F_B represents the body-fixed frame. The base frame of the manipulator is denoted as F_0 , and the joint frames are represented by F_i with $i = 1, 2, 3, 4$. The coordinate origin of F_B is denoted by point O , which is located at the CoM of the hex-rotor. Point S represents the CoM of the whole UAM system, and the position of S changes with the movement of the manipulator. $l_i (i = 1, 2, 3)$ represent the link length, and $\theta_i (i = 1, 2, 3)$ denote the joint angles. The definitions of the symbols are shown in Table 1. Then, through the linear and angular

Table 1

Symbols and definitions of the system.

Symbol	Definition
$m \in \mathbb{R}$	Mass of the aerial manipulator system
$\mathbf{p} = [x, y, z]^T \in \mathbb{R}^3$	Hex-rotor UAV position in F_I
$\mathbf{v} \in \mathbb{R}^3$	Hex-rotor UAV velocity in F_I
$\boldsymbol{\omega} \in \mathbb{R}^3$	Angular velocity of hex-rotor UAV in F_I
$R \in SO(3)$	Rotation matrix from F_B to F_I
$g \in \mathbb{R}$	Gravitational acceleration
$f \in \mathbb{R}$	Collective thrust generated by all rotating rotors
$\mathbf{M} \in \mathbb{R}^3$	Total moment generated by Hex-rotor UAV in F_B
$I \in \mathbb{R}^{3 \times 3}$	Inertia matrix of UAM referenced to O calculated in F_B
$m_m \in \mathbb{R}$	Mass of the manipulator
$m_i (i = 1, 2, 3) \in \mathbb{R}$	Mass of the link i
${}^B \mathbf{p}_{ci} \in \mathbb{R}^3$	Position of CoM of link i in F_B
$I_m \in \mathbb{R}^{3 \times 3}$	Inertia matrix of manipulator referenced to O calculated in F_B

momentum theorem of the mass point system, the dynamic model of the UAV equipped with a manipulator is described as follows:

$$\dot{\mathbf{p}} = \mathbf{v}, \quad (1)$$

$$m\dot{\mathbf{v}} = mg\mathbf{e}_3 - f\mathbf{Re}_3 + \underbrace{\mathbf{d}_{if1} + \mathbf{d}_{if2} + \mathbf{d}_{ef}}_{\mathbf{d}_f}, \quad (2)$$

$$\dot{R} = R\hat{\boldsymbol{\omega}}, \quad (3)$$

$$I\dot{\boldsymbol{\omega}} = \mathbf{M} - \boldsymbol{\omega} \times I\boldsymbol{\omega} + \underbrace{\mathbf{d}_{im1} + \mathbf{d}_{im2} + \mathbf{d}_{em}}_{\mathbf{d}_M}, \quad (4)$$

where $\mathbf{e}_3 = [0, 0, 1]^T$. The hat map $\hat{\cdot} : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ is defined by the relation that $\hat{\mathbf{x}}\mathbf{y} = \mathbf{x} \times \mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^3$. $\mathbf{d}_{ef} \in \mathbb{R}^3$ and $\mathbf{d}_{em} \in \mathbb{R}^3$ denote the external disturbances and the model uncertainties of force and moment, respectively. Among them, the external disturbances include wind, gust, ground effect, etc., and the model uncertainties include motor loss, battery voltage drop, air resistance, etc. As presented in Zhang et al. (2019), terms $\mathbf{d}_{if1} \in \mathbb{R}^3$, $\mathbf{d}_{if2} \in \mathbb{R}^3$, $\mathbf{d}_{im1} \in \mathbb{R}^3$ and $\mathbf{d}_{im2} \in \mathbb{R}^3$ describe the wrench disturbances, which are caused by the motion of the manipulator, that are

$$\mathbf{d}_{if1} = -mR(\boldsymbol{\omega} \times (\boldsymbol{\omega} \times {}^B \mathbf{r}_{os})), \quad (5)$$

$$\mathbf{d}_{if2} = -mR(\dot{\boldsymbol{\omega}} \times {}^B \mathbf{r}_{os} + {}^B \dot{\mathbf{r}}_{os} + 2\boldsymbol{\omega} \times {}^B \dot{\mathbf{r}}_{os}), \quad (6)$$

$$\mathbf{d}_{im1} = {}^B \mathbf{r}_{os} \times (f\mathbf{e}_3 - R^{-1}\mathbf{d}_{if1}), \quad (7)$$

$$\begin{aligned} \mathbf{d}_{im2} = & -{}^B \mathbf{r}_{os} \times R^{-1}\mathbf{d}_{if2} - \dot{I}_m \boldsymbol{\omega} \\ & - \frac{m^2}{m_m} ({}^B \mathbf{r}_{os} \times {}^B \dot{\mathbf{r}}_{os} + \boldsymbol{\omega} \times ({}^B \mathbf{r}_{os} \times {}^B \dot{\mathbf{r}}_{os})), \end{aligned} \quad (8)$$

where ${}^B \mathbf{r}_{os} \in \mathbb{R}^3$ is the vector from point O to S with respect to F_B , representing the CoM offset, and it can be calculated by ${}^B \mathbf{r}_{os} = f_{\theta}(\theta_i)(i = 1, 2, 3) = \frac{1}{m} \sum_{i=1}^3 m_i {}^B \mathbf{p}_{ci}$. It is worth noting that some terms cannot be directly measured by sensors, such as $\dot{\boldsymbol{\omega}}$, ${}^B \dot{\mathbf{r}}_{os}$, ${}^B \dot{\mathbf{r}}_{os}$ and $\dot{I}_m$, making it hard to compensate the contents in $\mathbf{d}_{if2}$ and $\mathbf{d}_{im2}$ while designing controller. Therefore, disturbances caused by the motion of the manipulator are divided into two parts: measurable $\mathbf{d}_{if1}$ and $\mathbf{d}_{im1}$, and unmeasurable $\mathbf{d}_{if2}$ and $\mathbf{d}_{im2}$, based on whether they contain the unmeasurable items mentioned above. $\mathbf{d}_f \in \mathbb{R}^3$ and $\mathbf{d}_M \in \mathbb{R}^3$ are the force uncertainty and the moment uncertainty, respectively.

Subsequently, define the state $\Xi = (\mathbf{p}, \mathbf{v}, R, \boldsymbol{\omega}) \in \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3) \times \mathbb{R}^3$, and its vector form $\xi \in \mathbb{R}^{18}$ is

$$\xi = [\mathbf{p}^T, \mathbf{v}^T, \text{vec}(R)^T, \boldsymbol{\omega}^T]^T, \quad (9)$$

where the vectorization operator $\text{vec}(R)$ represents arranging the columns of the matrix R into a column vector by stacking them vertically. Similarly, define the desired time-varying continuous state trajectory $\Xi_d = (\mathbf{p}_d, \mathbf{v}_d, R_d, \boldsymbol{\omega}_d) \in \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3) \times \mathbb{R}^3$, and its vector form $\xi_d = [\mathbf{p}_d^T, \mathbf{v}_d^T, \text{vec}(R_d)^T, \boldsymbol{\omega}_d^T]^T$. Then, the dynamic model (1)–(4) can be rewritten into a state-space form as follows:

$$\dot{\xi}(t) = F_u(\xi(t)) + G_u(R(t))[\mathbf{u}(t) + \boldsymbol{\gamma}_m(t, \xi(t))]$$

$$+ G_u^\perp(R(t))\boldsymbol{\gamma}_{um}(t, \xi(t)), \quad (10)$$

where $\mathbf{u} = [f, \mathbf{M}^T]^T \in \mathbb{R}^4$ denotes the control input generated by UAV. Considering the underactuation of the hex-rotor UAV, $\boldsymbol{\gamma}_m \in \mathbb{R}^4$ and $\boldsymbol{\gamma}_{um} \in \mathbb{R}^2$ denote the matched and unmatched uncertainties, respectively. In addition, both $\boldsymbol{\gamma}_m$ and $\boldsymbol{\gamma}_{um}$ are modeled as non-parametric uncertainties. The matched uncertainties $\boldsymbol{\gamma}_m$ can be divided into two parts: thrust uncertainty $\gamma_f \in \mathbb{R}$ and moment uncertainty $\boldsymbol{\gamma}_M \in \mathbb{R}^3$, i.e., $\boldsymbol{\gamma}_m = [\gamma_f, \boldsymbol{\gamma}_M^T]^T$. Specifically, the relationship between $\boldsymbol{\gamma}_m, \boldsymbol{\gamma}_{um}$ and $\mathbf{d}_f, \mathbf{d}_M$ can be described as

$$\mathbf{d}_f = [\mathbf{Re}_1, \mathbf{Re}_2]\boldsymbol{\gamma}_{um} - \gamma_f \mathbf{Re}_3, \quad (11)$$

$$\mathbf{d}_M = \boldsymbol{\gamma}_M + \gamma_f {}^B \mathbf{r}_{os} \times \mathbf{e}_3, \quad (12)$$

where $\mathbf{e}_1 = [1, 0, 0]^T$ and $\mathbf{e}_2 = [0, 1, 0]^T$. The form of $F_u(\xi) \in \mathbb{R}^{18 \times 1}$, $G_u(R) \in \mathbb{R}^{18 \times 4}$ and $G_u^\perp(R) \in \mathbb{R}^{18 \times 2}$ are defined by

$$F_u(\xi) = \begin{bmatrix} \mathbf{v} \\ g\mathbf{e}_3 - R(\boldsymbol{\omega} \times (\boldsymbol{\omega} \times {}^B \mathbf{r}_{os})) \\ \text{vec}(R\hat{\boldsymbol{\omega}}) \\ I^{-1}({}^B \mathbf{r}_{os} \times (m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times {}^B \mathbf{r}_{os})) - \boldsymbol{\omega} \times I\boldsymbol{\omega}) \end{bmatrix},$$

$$G_u(R) = \begin{bmatrix} 0_{3 \times 1} & 0_{3 \times 3} \\ -\frac{1}{m}\mathbf{Re}_3 & 0_{3 \times 3} \\ 0_{9 \times 1} & 0_{9 \times 3} \\ I^{-1}({}^B \mathbf{r}_{os} \times \mathbf{e}_3) & I^{-1} \end{bmatrix}, G_u^\perp(R) = \begin{bmatrix} 0_{3 \times 1} & 0_{3 \times 1} \\ \frac{1}{m}\mathbf{Re}_1 & \frac{1}{m}\mathbf{Re}_2 \\ 0_{9 \times 1} & 0_{9 \times 1} \\ 0_{3 \times 1} & 0_{3 \times 1} \end{bmatrix},$$

where the columns of $G_u^\perp(R)$ is perpendicular to those of $G_u(R)$.

Remark 1. The core sensors of traditional inertial measurement units (IMUs) include accelerometers and gyroscopes, which are used to measure linear acceleration and angular velocity, respectively. For angular acceleration, the measurement data must be processed and computed to indirectly obtain the relevant information. Since angular acceleration is difficult to measure directly and accurately, algorithms based on Kalman filtering can be employed to reduce the impact of noise and errors, or the sensor structure can be designed and improved to enable direct measurement of angular acceleration. This paper focuses on compensating for uncertainties that contain hard-to-accurately-measure components, based on existing hardware.

3. Control of the aerial manipulator system

3.1. Control objective

Define the Euclidean norm of the state error as

$$D(\xi, \xi_d) = \left\| \begin{bmatrix} \mathbf{e}_p^T, \mathbf{e}_v^T, \mathbf{e}_R^T, \mathbf{e}_\omega^T \end{bmatrix}^T \right\|, \quad (13)$$

where $\mathbf{e}_p = \mathbf{p} - \mathbf{p}_d$ and $\mathbf{e}_v = \mathbf{v} - \mathbf{v}_d$ are the position and velocity errors, respectively. $\mathbf{e}_R = \frac{1}{2}(R_d^T R - R^T R_d)^\vee$ and $\mathbf{e}_\omega = \boldsymbol{\omega} - R^T R_d \boldsymbol{\omega}_d$ represent the attitude and angular velocity errors, respectively, where the vee map $(\cdot)^\vee$ is the inverse operation of the hat map $\hat{\cdot}$. Define $\mathcal{O}(\xi_d(t), \rho)$ as a set centered at ξ_d with radius ρ , that is, $\mathcal{O}(\xi_d(t), \rho) = \{\xi \in \mathbb{R}^{18} | D(\xi, \xi_d) \leq \rho\}$ where ρ is a positive scalar. Then, a tube centered around the desired trajectory for all $t \geq 0$, is described as

$$\mathcal{T}(\rho) = \cup_{t \geq 0} \mathcal{O}(\xi_d(t), \rho). \quad (14)$$

The control objective of this paper is to design the control input $\mathbf{u}$ to drive the state ξ of the uncertain system (10) always within the set $\mathcal{O}(\xi_d(t), \rho)$, that is

$$\xi(t) \in \mathcal{O}(\xi_d(t), \rho), \forall t \geq 0. \quad (15)$$

Considering the physical limitations of practical systems and without loss of generality (Chen, Liang et al., 2022; Cheng et al., 2022; Hovakimyan & Cao, 2010; Lakshmanan et al., 2020; Liang et al., 2025; Lyu et al., 2025; Wu et al., 2023; Zhao et al., 2024; Zuo & Ru, 2014), the following assumptions regarding uncertainties and system functions are made:

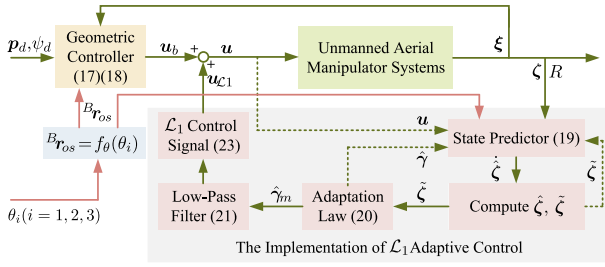


Fig. 2. The control block diagram. The gray area represents the implementation of $\mathcal{L}_1$ adaptive control, dashed line indicates the transmission of the signal from the previous time step.

Assumption 1. The matched and unmatched uncertainties $\gamma(t, \xi(t)) = [\gamma_m(t, \xi(t))^T, \gamma_{um}(t, \xi(t))^T]^T$ are bounded and are Lipschitz continuous with respect to both ξ and t , i.e.,

$$\begin{aligned} \|\gamma(t, \xi(t))\| &\leq \Delta_\gamma, \\ \|\gamma_m(t, \xi(t))\| &\leq \Delta_{\gamma_m}, \\ \|\gamma_{um}(t, \xi(t))\| &\leq \Delta_{\gamma_{um}}, \\ \|\gamma(t_1, \xi(t_1)) - \gamma(t_2, \xi(t_2))\| &\leq L_{\gamma_\xi} \|\xi(t_1) - \xi(t_2)\| \\ &\quad + L_{\gamma_t} |t_1 - t_2|, \\ \|\gamma_m(t_1, \xi(t_1)) - \gamma_m(t_2, \xi(t_2))\| &\leq L_{\gamma_{m\xi}} \|\xi(t_1) - \xi(t_2)\| \\ &\quad + L_{\gamma_{mt}} |t_1 - t_2|, \end{aligned}$$

for all $\xi \in \mathcal{T}(\rho)$, where $\Delta_\gamma, \Delta_{\gamma_m}, \Delta_{\gamma_{um}}$ are constants. L_{γ_ξ} and L_{γ_t} are the Lipschitz constants of $\gamma(t, \xi(t))$ with respect to the state ξ and time t , respectively. Similarly, $L_{\gamma_{m\xi}}$ and $L_{\gamma_{mt}}$ denote the Lipschitz constants of $\gamma_m(t, \xi(t))$ with respect to ξ and t , respectively.

Assumption 2. There exist positive constant L_G such that

$$\|\bar{G}_u(t_1) - \bar{G}_u(t_2)\| \leq L_G |t_1 - t_2|,$$

for all $t \geq 0$, where $\bar{G}_u(R(t)) = [G_u(R(t)), G_u^\perp(R(t))]$.

Remark 2. For disturbances associated with state acceleration, the corresponding Lipschitz constants in **Assumption 1** may be notably large, making the selection of controller parameters in the subsequent stability analysis challenging. Hence, the proposed method has certain limitations when applied to scenarios involving high acceleration.

3.2. Controller design

To handle the uncertainties d_f and d_M that contain unmeasurable terms and unknown external disturbances, an $\mathcal{L}_1$ adaptive augmentation control method is proposed in this subsection. The control block diagram of the proposed method is shown in Fig. 2.

Specifically, the implemented $\mathcal{L}_1$ adaptive controller is primarily composed of three components: a state predictor, an adaptive law, and a low-pass filter (LPF). The state predictor is employed to predict the evolution of the system's state, where the prediction error is used by the adaptive law to update the disturbance estimation. Considering that the uncertainties only appear in (2) and (4), define the reduced state $\zeta = [v^T, \omega^T]^T \in \mathbb{R}^6$, which is directly affected by the uncertainties. Furthermore, the state-space equation of (10) with partial state ζ is described as follows:

$$\begin{aligned} \dot{\zeta}(t) &= F(\zeta(t)) + G(R(t))[u(t) + \gamma_m(t, \zeta(t))] \\ &\quad + G^\perp(R(t))\gamma_{um}(t, \zeta(t)), \end{aligned} \quad (16)$$

where

$$F(\zeta) = \begin{bmatrix} g e_3 - R(\omega \times (\omega \times {}^B r_{os})) \\ I^{-1}({}^B r_{os} \times (m \omega \times (\omega \times {}^B r_{os})) - \omega \times I \omega) \end{bmatrix},$$

$$G(R) = \begin{bmatrix} -\frac{1}{m} R e_3 & 0_{3 \times 3} \\ I^{-1}({}^B r_{os} \times e_3) & I^{-1} \end{bmatrix}, G^\perp(R) = \begin{bmatrix} \frac{1}{m} R e_1 & \frac{1}{m} R e_2 \\ 0_{3 \times 1} & 0_{3 \times 1} \end{bmatrix}.$$

The control input u can be divided as $u = u_b + u_{\mathcal{L}_1}$, where $u_b = [f_b, M_b^T]^T \in \mathbb{R}^4$ denotes the basic geometric control signal, and $u_{\mathcal{L}_1} = [f_{\mathcal{L}_1}, M_{\mathcal{L}_1}^T]^T \in \mathbb{R}^4$ represents the $\mathcal{L}_1$ control signal. Inspired by the CoM offset compensation control method in Zhang et al. (2018), for the uncertainty-free dynamics with $d_f = d_M = 0$ in (1)–(4), the geometric controller u_b is designed as

$$f_b = -F_d^T R e_3, \quad (17)$$

$$\begin{aligned} M_b &= -K_R e_R - K_\omega e_\omega - {}^B r_{os} \times (f_b e_3 + m \omega \times (\omega \times {}^B r_{os})) \\ &\quad + \omega \times I \omega - I(\hat{\omega} R^T R_d \omega_d - R^T R_d \hat{\omega}_d), \end{aligned} \quad (18)$$

where the virtual control force $F_d \in \mathbb{R}^3$ is calculated by $F_d = -K_p e_p - K_v e_v - m g e_3 + m R(\omega \times (\omega \times {}^B r_{os})) + m \ddot{p}_d$, wherein $K_p, K_v \in \mathbb{R}^{3 \times 3}$ are positive-definite diagonal control matrices. K_R and K_ω are yet-to-be-determined positive constant gains. $R_d = [r_{yd} \times r_{zd}, r_{yd}, r_{zd}] \in SO(3)$ is the desired attitude, where the desired unit direction vector $r_{zd} \in \mathbb{R}^3$ can be obtained by $r_{zd} = R_d e_3 = -\frac{F_d}{\|F_d\|}$, $r_{yd} = \frac{r_{zd} \times r_{id}}{\|r_{zd} \times r_{id}\|}$ represents the second row of the desired rotation matrix, where $r_{id} \in \mathbb{R}^3$ is an arbitrary vector that is not parallel to r_{zd} . The goal of the baseline geometric controller is to have the system along the desired trajectory p_d and yaw ψ_d .

Subsequently, define the state predictor as:

$$\begin{aligned} \dot{\zeta}(t) &= F(\zeta(t)) + G(R(t))[u_b(t) + u_{\mathcal{L}_1}(t) + \hat{\gamma}_m(t, \zeta(t))] \\ &\quad + G^\perp(R(t))\hat{\gamma}_{um}(t, \zeta(t)) + A_s \tilde{\zeta}(t), \end{aligned} \quad (19)$$

where $\tilde{\zeta} = \hat{\zeta} - \zeta$ represents the prediction error, $\hat{\zeta}$ is the predicted state of the predictor, A_s is a Hurwitz matrix representing the adaptation gain, $\hat{\gamma}_m$ and $\hat{\gamma}_{um}$ are the current adaptive estimates of uncertainties γ_m and γ_{um} , respectively.

Following that, in accordance with the design of the piecewise-constant (PWC) adaptive law in (Hovakimyan & Cao, 2010, Section 3.3), the estimated uncertainty is calculated such that for $t \in [iT, (i+1)T)$,

$$\hat{\gamma}(t) = \hat{\gamma}(iT) = \begin{bmatrix} \hat{\gamma}_m(iT) \\ \hat{\gamma}_{um}(iT) \end{bmatrix} = -\bar{G}^{-1}(iT) \Phi^{-1} \mu(iT), \quad (20)$$

where T represents the adaptation sampling time, $\Phi = A_s^{-1}[e^{(A_s T)} - E_{6 \times 6}]$, $E_{n \times n}$ represents an $n \times n$ identity matrix. $\bar{G}(iT) = [G(R(iT)), G^\perp(R(iT))]$ and $\mu(iT) = e^{A_s T} \tilde{\zeta}(iT)$ are calculated at time step i .

Due to the introduction of high-frequency components in the estimation with high adaptive gains, a low-pass filter will be adopted. The role of the low-pass filter here is to extract the low-frequency components of the signal, suppress the interference of high-frequency components introduced by uncertainty estimation, and thus prevent high-frequency components from entering the control channel. The LPF is designed as a first-order filter with a bandwidth of ω_c , taking the following form:

$$C(s) = \frac{\omega_c}{s + \omega_c}, \quad (21)$$

then, the $\mathcal{L}_1$ control law is formulated

$$u_{\mathcal{L}_1}(s) = -C(s)\hat{\gamma}_m(s), \quad (22)$$

where the signals in (22) are placed in the Laplacian domain. Furthermore, the control law is practically implemented as follows:

$$u_{\mathcal{L}_1,k} = e^{-\omega_c T} u_{\mathcal{L}_1,k-1} - (1 - e^{-\omega_c T}) \hat{\gamma}_{m,k}, \quad (23)$$

where $k-1$ and k are previous and current steps, respectively.

Remark 3. By substituting (19) into (16), one has $\dot{\zeta}(t) = A_s \tilde{\zeta}(t) + \bar{G}(t)[\hat{\gamma}(t) - \gamma(t)]$, then define a Lyapunov candidate function as $V_\zeta = \frac{1}{2} \tilde{\zeta}^T \tilde{\zeta}$, its derivative is

$$\dot{V}_\zeta = \dot{\zeta}^T \tilde{\zeta} = (A_s \tilde{\zeta})^T \tilde{\zeta} + [\bar{G}(\hat{\gamma} - \gamma)]^T \tilde{\zeta},$$

$$= -A_s(e^{(A_s T)} - E_{6 \times 6})^{-1} \tilde{\xi}^T \tilde{\xi} - \tilde{G} \gamma^T \tilde{\xi}.$$

For the user-defined Hurwitz matrix A_s and sampling interval T , it is generally possible to choose a sufficiently small T and an appropriate A_s , such that $e^{(A_s T)}$ approach and slightly smaller than the identity matrix $E_{6 \times 6}$, then $A_s(e^{(A_s T)} - E_{6 \times 6})^{-1} \tilde{\xi}^T \tilde{\xi}$ is greater than $\tilde{G} \gamma^T \tilde{\xi}$, ensuring that $\dot{V}_{\tilde{\xi}}$ is negative definite. Consequently, the stability of the state predictor has been proven through Lyapunov-based analysis.

Remark 4. Since $\det(R) = 1$, ${}^B r_{os}$ and I are bounded, one can conclude that the functions $G(R(t))$, $\tilde{G}(R(t))$, $\tilde{G}^{-1}(R(t))$, $G_u(R(t))$, $\tilde{G}_u(R(t))$ are bounded and satisfy

$$\begin{aligned} \|G(R(t))\| &\leq \Delta_G, & \|\tilde{G}(R(t))\| &\leq \Delta_{\tilde{G}}, \\ \|\tilde{G}^{-1}(R(t))\| &\leq \Delta_{\tilde{G}^{-1}}, & \|G_u(R(t))\| &\leq \Delta_{G_u}, \\ \|\tilde{G}_u(R(t))\| &\leq \Delta_{\tilde{G}_u}, \end{aligned}$$

for all $t \geq 0$, where $\Delta_G, \Delta_{\tilde{G}}, \Delta_{\tilde{G}^{-1}}, \Delta_{G_u}, \Delta_{\tilde{G}_u}$ are constants.

4. Stability analysis

In this section, the stability of the uncertainty-free system based on the controller u_b is first demonstrated. Based on this, the performance of the proposed method with consideration of the influence caused by disturbances and uncertainties is analyzed.

Before presenting the proof, the definitions of the following function and matrices are introduced. Define the error function in $SO(3)$ as

$$\Psi(R, R_d) = \frac{1}{2} \text{tr}(E_{3 \times 3} - R_d^T R). \quad (24)$$

Matrices $W_1, W_2, W_{12} \in \mathbb{R}^{2 \times 2}$ and $W \in \mathbb{R}^{4 \times 4}$ are given by

$$\begin{aligned} W_1 &= \begin{bmatrix} \frac{c_1 \lambda_{\min}(K_p)}{m} (1 - \alpha) & w_0 \\ w_0 & \lambda_{\min}(K_v) (1 - \alpha) - c_1 \end{bmatrix}, \\ W_2 &= \begin{bmatrix} \frac{c_2 K_R}{\lambda_{\max}(I)} & -\frac{1}{2} (\frac{c_2 K_{\omega} + K_R}{\lambda_{\min}(I)} - c_0 K_R) \\ -\frac{1}{2} (\frac{c_2 K_{\omega} + K_R}{\lambda_{\min}(I)} - c_0 K_R) & \frac{K_{\omega}}{\lambda_{\max}(I)} - c_2 \end{bmatrix}, \\ W_{12} &= \begin{bmatrix} \frac{c_1}{m} B & 0 \\ B & 0 \end{bmatrix}, \quad W = \begin{bmatrix} W_1 & -\frac{W_{12}}{2} \\ -\frac{W_{12}^T}{2} & W_2 \end{bmatrix}, \end{aligned}$$

where $c_0, c_1, c_2 \in \mathbb{R}$ are positive constants, ψ_1 is a fixed constant such that $\Psi(R(0), R_d(0)) < \psi_1 < 1$, $\alpha = \sqrt{\psi_1(2 - \psi_1)}$, B is a given positive constant such that $B > \left\| -m g e_3 + m \ddot{p}_d + m R(\omega \times (\omega \times {}^B r_{os})) \right\|$, $w_0 = -\frac{c_1 \lambda_{\max}(K_v)}{2m} (1 + \alpha) - \frac{\lambda_{\max}(K_p) \alpha}{2}$, $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$ denote the maximum and minimum eigenvalues of the square matrix A , respectively. Define $M_{11}, M_{12}, M_{21}, M_{22} \in \mathbb{R}^{2 \times 2}$ as follows:

$$\begin{aligned} M_{11} &= \frac{1}{2} \begin{bmatrix} \lambda_{\min}(K_p) & -c_1 \\ -c_1 & m \end{bmatrix}, & M_{12} &= \frac{1}{2} \begin{bmatrix} \lambda_{\max}(K_p) & c_1 \\ c_1 & m \end{bmatrix}, \\ M_{21} &= \frac{1}{2} \begin{bmatrix} c_0 K_R & -c_2 \\ -c_2 & 1 \end{bmatrix}, & M_{22} &= \frac{1}{2} \begin{bmatrix} \frac{2c_0 K_R}{2 - \psi_1} & c_2 \\ c_2 & 1 \end{bmatrix}. \end{aligned}$$

Proposition 1. Considering the thrust force f_b and moment M_b designed for the uncertainty-free system, i.e., the case that $d_f = d_M = 0$ in (1)–(4), suppose that the initial conditions satisfy

$$\Psi(R(0), R_d(0)) < \psi_1 < 1. \quad (25)$$

Choose appropriate positive constants $K_R, K_{\omega}, c_0, c_1, c_2$ to ensure that matrices $W, W_1, W_2, M_{11}, M_{21}, M_{12}, M_{22}$ are positive definite. Then, the equilibrium of the closed-loop system is exponentially stable. The attraction region is characterized by (25) and

$$\|e_{\omega}(0)\|^2 < \frac{2}{\lambda_{\max}(I)} K_R (\psi_1 - \Psi(R(0), R_d(0))). \quad (26)$$

Proof. The proof of Proposition 1 is divided into two parts: the exponential stability of attitude dynamics in the sublevel set $L_1 = \{R_d, R \in SO(3) \mid \Psi(R, R_d) < 2\}$ (Step 1), and the exponential stability of the closed-loop uncertainty-free system dynamics in a smaller sublevel set $L_2 = \{R_d, R \in SO(3) \mid \Psi(R, R_d) < 1\}$ (Step 2).

Step 1: According to the definition of e_{ω} , the time derivative of e_R and e_{ω} are $\dot{e}_R = \frac{1}{2} (R_d^T R \dot{e}_{\omega} + \dot{e}_{\omega} R^T R_d)^V$ and $\dot{e}_{\omega} = \dot{\omega} + \dot{\omega}^T R_d \omega_d - R^T R_d \dot{\omega}_d$, respectively. Together with (4) and (18) leads to

$$I \dot{e}_{\omega} = -K_R e_R - K_{\omega} e_{\omega}. \quad (27)$$

Next, define a Lyapunov candidate function as

$$V_1 = \frac{1}{2} e_{\omega}^T e_{\omega} + c_0 K_R \Psi(R, R_d) + c_2 e_R^T e_{\omega}. \quad (28)$$

Following the proof given in Lee et al. (2010, Proposition 1), one can conclude that the zero equilibrium of the attitude tracking error e_R and e_{ω} is exponentially stable.

Step 2: Considering the definition of e_v , the time derivative of e_v is given by:

$$m \dot{e}_v = m g e_3 - f_b R e_3 - m R(\omega \times (\omega \times {}^B r_{os})) - m \ddot{p}_d. \quad (29)$$

Substituting (17) into (29), leads to

$$m \dot{e}_v = -K_p e_p - K_v e_v - \Delta, \quad (30)$$

where $\Delta = \frac{f_b}{e_3^T R_d^T R e_3} [(e_3^T R_d^T R e_3) R e_3 - R_d e_3]$ is an auxiliary term. Then, define a Lyapunov candidate function as

$$V_2 = \frac{1}{2} e_p^T K_p e_p + \frac{1}{2} m e_v^T e_v + c_1 e_p^T e_v. \quad (31)$$

Noting that by an extension of the proof provided in Lee et al. (2010, Proposition 3), let $V = V_1 + V_2$, one can get

$$\kappa D(\xi, \xi_d)^2 \leq V \leq \bar{\kappa} D(\xi, \xi_d)^2, \quad (32)$$

$$\dot{V} \leq -\beta V, \quad (33)$$

where

$$\begin{aligned} \kappa &= \min(\lambda_{\min}(M_{11}), \lambda_{\min}(M_{21})), \\ \bar{\kappa} &= \max(\lambda_{\max}(M_{12}), \lambda_{\max}(M_{22})), \\ \beta &\leq \frac{\lambda_{\min}(W)}{\lambda_{\max}(\text{diag}(M_{12}, M_{22}))}. \end{aligned}$$

Hereto, the equilibrium of the closed-loop system is exponentially stable, and the proof of Proposition 1 is completed. $\square$

Proposition 2. Define

$$\begin{aligned} P_1(t) &= \begin{bmatrix} -(R e_3)^T (\frac{c_1}{m} e_p + e_v) \\ (I^{-1})^T (e_{\omega} + c_2 e_R) \end{bmatrix}^T \\ &\quad + \begin{bmatrix} ({}^B r_{os} \times e_3)^T (I^{-1})^T (e_{\omega} + c_2 e_R) \\ 0 \end{bmatrix}^T, \end{aligned} \quad (34)$$

$$P_2(t) = \frac{c_1}{m} e_p^T + e_v^T, \quad (35)$$

there exist constants $c_3, c_4, \Delta_F, \Delta_{u_b}, \Delta_{u_{L1}}$ such that $P_1, P_2, F_u(\xi(t)), u_b, u_{L1}$ are bounded by

$$\begin{cases} \|P_1\| \leq c_3 \rho, \|P_2\| \leq c_4 \rho, \|F_u(\xi(t))\| \leq \Delta_F \\ \|u_b\| \leq \Delta_{u_b}, \|u_{L1}\| \leq \Delta_{u_{L1}} \end{cases} \quad (36)$$

for all $\xi \in \mathcal{O}(\xi_d(t), \rho)$ and $t \geq 0$.

Proof. The proof in Wu et al. (2023, Proposition 3) demonstrates that $\|P_2\|, \|F_u(\xi(t))\|, \|u_b\|, \|u_{L1}\|$ in (36) are bounded. In comparison to the original proof, the introduction of $({}^B r_{os} \times e_3)^T (I^{-1})^T (e_{\omega} + c_2 e_R)$ in P_1 originates from the term ${}^B r_{os} \times f_{e_3}$ in the proposed controller. Specifically, based on the definition of $\xi \in \mathcal{O}(\xi_d(t), \rho)$, it can be deduced that $\|e_p\|, \|e_v\|, \|e_R\|, \|e_{\omega}\|$ have upper bound ρ . Considering that c_2 is constant, $\|R e_3\| = 1$, ${}^B r_{os} \times e_3$ and I^{-1} are bounded, it follows that the bound of $\|P_1\|$ in (36) hold. $\square$

Next, for the convenience of subsequent analysis, define

$$\begin{aligned}\phi_1 &= \Delta_F + \Delta_{\bar{G}} \Delta_\gamma + \Delta_{G_u} \left(\Delta_{u_b} + \Delta_{u_{L1}} \right), \\ \varsigma_1(\omega_c) &= \frac{\Delta_{\gamma_m}}{|\beta - \omega_c|} + \frac{L_{\gamma_{m_t}} + L_{\gamma_{m_\xi}} \phi_1}{\beta \omega_c}, \\ \varsigma_2(A_s) &= 2\sqrt{6} \Delta_{\bar{G}} \Delta_{\bar{G}^{-1}} \left(\phi_1 L_{\gamma_\xi} + L_{\gamma_t} \right) + 2\sqrt{6} \Delta_{\bar{G}^{-1}} L_G \Delta_{u_{L1}} \\ &\quad + \sqrt{6} \Delta_{\bar{G}^{-1}} \left(2L_G + |\lambda_{\min}(A_s)| \Delta_{\bar{G}} \right) \Delta_\gamma, \\ \varsigma_3(\omega_c) &= \Delta_\gamma \omega_c, \\ \varsigma_4(A_s, \omega_c) &= \varsigma_2(A_s) + \varsigma_3(\omega_c), \\ \rho &= D(\xi(0), \xi_d(0)) \sqrt{\kappa/\kappa} + \epsilon,\end{aligned}$$

where ϵ represents an arbitrary small value. The first-order low-pass filter $C(s)$ in the form of (21) is employed, and its sampling time T and bandwidth ω_c need to satisfy the following conditions:

$$\kappa \rho^2 > c_3 \rho \varsigma_1(\omega_c) + c_4 \rho \Delta_{\gamma_{um}} + V_0, \quad (37)$$

$$T \leq \frac{\kappa \rho^2 - c_3 \rho \varsigma_1(\omega_c) - c_4 \rho \Delta_{\gamma_{um}} - V_0}{\varsigma_4(A_s, \omega_c)}, \quad (38)$$

where $V_0 = V(e_p(0), e_v(0), e_R(0), e_\omega(0))$. It is worth noting that as ω_c increases, $\varsigma_1(\omega_c)$ converges to 0, and if $\Delta_{\gamma_{um}} \leq \frac{\kappa \rho^2 - V_0 - \epsilon}{c_4 \rho}$ for all $\xi \in \mathcal{T}(\rho)$ is satisfied, conditions (37) and (38) can always be satisfied by selecting a sufficiently small T and a sufficiently large ω_c .

Subsequently, the main theorem of this paper is introduced.

Theorem 1. Under the conditions in (37) and (38), utilizing the proposed controller u , the closed-loop system state satisfy:

$$\xi(t) \in \mathcal{O}(\xi_d(t), \rho), \quad \forall t \geq 0. \quad (39)$$

Furthermore, the state is uniformly ultimately bounded, that is

$$\xi(t) \in \mathcal{O}(\xi_d(t), \mu(\omega_c, T, t_1)) \subset \mathcal{O}(\xi_d(t), \rho), \quad \forall t \geq t_1 > 0,$$

for any $t_1 > 0$, where the ultimate bound is given by

$$\mu(\omega_c, T, t_1) = \sqrt{\frac{e^{-\beta t_1} V_0 + c_3 \rho \varsigma_1(\omega_c) + \varsigma_4(A_s, \omega_c) T + c_4 \rho \Delta_{\gamma_{um}}}{\kappa}}.$$

Proof. Firstly, considering the system dynamics with uncertainty (1)–(4), together with (11)–(12) and (29)–(30), $m\dot{e}_v$ can be rewritten as

$$\begin{aligned}m\dot{e}_v &= -K_p e_p - K_v e_v - \Delta - (f_{L1} + \gamma_f) Re_3 \\ &\quad + [Re_1, Re_2] \gamma_{um}.\end{aligned} \quad (40)$$

Similarly, with the result in (27), one can conclude that

$$I\dot{e}_\omega = -K_R e_R - K_\omega e_\omega + M_{L1} + \gamma_M + (f_{L1} + \gamma_f)^B r_{os} \times e_3. \quad (41)$$

Subsequently, substituting (40) and (41) into the time derivative of V , and combining with (33), it can be deduced that

$$\begin{aligned}\dot{V} &\leq -\beta V + \left(\frac{c_1}{m} e_p^\top + e_v^\top \right) ([Re_1, Re_2] \gamma_{um}) \\ &\quad + \left((e_\omega^\top + c_2 e_R^\top) I^{-1} \right) (B r_{os} \times e_3) (f_{L1} + \gamma_f) \\ &\quad + \left[-(Re_3)^\top \left(\frac{c_1}{m} e_p + e_v \right) \right]^\top \begin{bmatrix} f_{L1} + \gamma_f \\ M_{L1} + \gamma_M \end{bmatrix}.\end{aligned} \quad (42)$$

Then, define

$$\begin{cases} \eta_m(s) = C(s) \mathcal{L}[\gamma_m(t, \xi(t))] \\ \hat{\eta}_m(s) = C(s) \mathcal{L}[\hat{\gamma}_m(t, \xi(t))] \end{cases}. \quad (43)$$

By substituting (22) and (43) into (42), one can conclude that

$$\begin{aligned}\dot{V} &\leq -\beta V + P_1(\gamma_m - \eta_m) + P_1(\eta_m - \hat{\eta}_m) \\ &\quad + P_2([Re_1, Re_2] \gamma_{um}),\end{aligned} \quad (44)$$

where the definitions of P_1 and P_2 can be found in (34) and (35).

Subsequently, utilizing contradiction to prove this theorem. Under Assumption 1 and conditions in (37) and (38), assuming that $D(\xi(t), \xi_d(t)) \geq \rho$ for some $t > 0$. In other words, considering $D(\xi(0), \xi_d(0)) < \rho$, there exists $\tau^* > 0$ such that

$$\begin{cases} D(\xi(t), \xi_d(t)) < \rho, & t \in [0, \tau^*) \\ D(\xi(\tau^*), \xi_d(\tau^*)) = \rho \end{cases}. \quad (45)$$

According to the differential inequality in (44), it can be derived that

$$V(t) \leq e^{-\beta t} V_0 + \Phi_1(t) + \Phi_2(t) + \Phi_3(t), \quad (46)$$

where

$$\Phi_1(t) = \int_0^t e^{-\beta(t-\tau)} P_1(\tau) (\gamma_m(\tau) - \eta_m(\tau)) d\tau,$$

$$\Phi_2(t) = \int_0^t e^{-\beta(t-\tau)} P_1(\tau) \hat{\eta}_m(\tau) d\tau,$$

$$\Phi_3(t) = \int_0^t e^{-\beta(t-\tau)} P_2([Re_1, Re_2] \gamma_{um}) d\tau,$$

and $\hat{\eta}_m(t) = \eta_m(t) - \hat{\eta}_m(t)$.

Summarizing the conclusions in Wu et al. (2023, Theorem 1), the bounds of $\|\Phi_1\|$, $\|\Phi_2\|$ and $\|\Phi_3\|$ can be concluded as follows:

$$\|\Phi_1\| \leq c_3 \rho \left(\frac{\Delta_{\gamma_m}}{|\beta - \omega_c|} + \frac{L_{\gamma_{m_t}} + L_{\gamma_{m_\xi}} \phi_1}{\beta \omega_c} \right) = c_3 \rho \varsigma_1(\omega_c), \quad (47)$$

$$\|\Phi_2\| \leq \frac{c_3 \rho}{\beta} (\varsigma_3(\omega_c) + \varsigma_2(A_s)) T = \varsigma_4(A_s, \omega_c) T, \quad (48)$$

$$\|\Phi_3\| \leq c_4 \rho \Delta_{\gamma_{um}}. \quad (49)$$

Then, substituting (47)–(49) into (46), one has

$$V \leq e^{-\beta t} V_0 + c_3 \rho \varsigma_1(\omega_c) + \varsigma_4(A_s, \omega_c) T + c_4 \rho \Delta_{\gamma_{um}}. \quad (50)$$

Considering the previous assumption of $D(\xi(\tau^*), \xi_d(\tau^*)) = \rho$ and the result in (32), one can conclude that

$$\kappa \rho^2 < V_0 + c_3 \rho \varsigma_1(\omega_c) + \varsigma_4(A_s, \omega_c) T + c_4 \rho \Delta_{\gamma_{um}}, \quad (51)$$

thus

$$T > \frac{\kappa \rho^2 - c_3 \rho \varsigma_1(\omega_c) - c_4 \rho \Delta_{\gamma_{um}} - V_0}{\varsigma_4(A_s, \omega_c)}, \quad (52)$$

which is contradictive with the conditions in (37) and (38). Therefore, for all $t \geq 0$, one has

$$D(\xi(t), \xi_d(t)) \leq \rho. \quad (53)$$

Furthermore, summarizing the conclusions of (32) and (50), the ultimate bound can be obtained, i.e., for any $t_1 > 0$,

$$\begin{aligned}D(\xi(t), \xi_d(t)) &< \mu(\omega_c, T, t_1) = \\ &\sqrt{\frac{e^{-\beta t_1} V_0 + c_3 \rho \varsigma_1(\omega_c) + \varsigma_4(A_s, \omega_c) T + c_4 \rho \Delta_{\gamma_{um}}}{\kappa}}\end{aligned} \quad (54)$$

holds for all $t \geq t_1 > 0$. Finally, combining with (53) and (54), the proof of Theorem 1 is completed. $\square$

5. Hardware experiments

To validate the performance of the proposed $\mathcal{L}_1$ adaptive augmentation control algorithm for the UAM system, three groups of hardware experiments are conducted in this section.

As illustrated in Fig. 3, the experiments are performed on a self-built testbed. There are 8 cameras and a computer in the Qualisys motion capture system to identify markers on the hexacopter, measure the state signals of the UAM system, and transmit the signals to the ground station via a Local Area Network (LAN). The corresponding control signals are computed in the ground station, which are then transmitted to the onboard Raspberry Pi via WiFi with 5G frequency band. Subsequently, control commands are sent to PixHawk and Dynamixel to

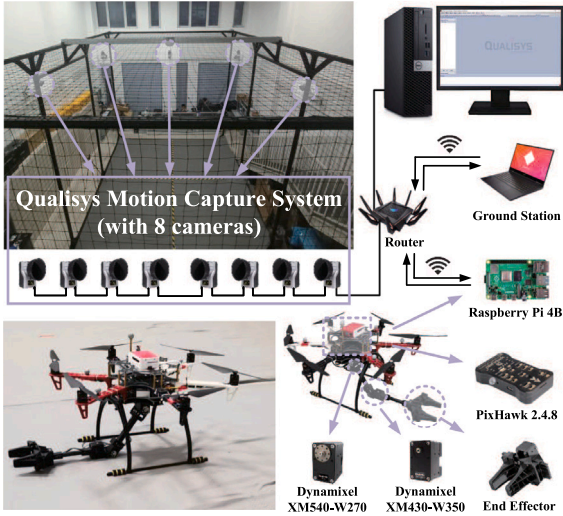


Fig. 3. The experiment testbed.

control the unmanned aerial manipulator system. The hexacopter, with a wingspan of 550 mm, is powered by Tmotor AIR2216-920kv motors. The manipulator features Dynamixel XM430-W350 and XM540-W270 servo motors, with 3D-printed linkages and an OpenManipulator RM-X52 gripper as the end effector, where θ_4 corresponds to the opening degree of the gripper. It is noteworthy that both the ground station and the onboard Raspberry Pi run the 64-bit Ubuntu 18.04 operating system, the data interactions in the project are supported by the communication protocols of the Robot Operating System (ROS).

The physical parameters of the testbed are as follows:

$$m = 3.01 \text{ kg}, g = 9.8 \text{ m/s}^2,$$

$$l_1 = 0.17 \text{ m}, l_2 = 0.05 \text{ m}, l_3 = 0.25 \text{ m}.$$

Three groups of tests are conducted to verify the effectiveness of the proposed control method, including the cases with manipulator swinging, trajectory tracking and pick-and-release task. Inspired by the control method with CoM offset compensation designed in Zhang et al. (2018), the baseline control method $u_b = [-F_d^T R e_3, \mathbf{M}_b^T]^T$ without to-be-estimated disturbances is as follows:

$$F_d = -K_p e_p - K_v e_v - m g e_3 + m R (\omega \times (\omega \times {}^B r_{os})) + m \dot{p}_d,$$

$$\mathbf{M}_b = -K_R e_R - K_\omega e_\omega - {}^B r_{os} \times (f_b e_3 + m \omega \times (\omega \times {}^B r_{os})) + \omega \times I \omega - I (\dot{\omega} R^T R d_\omega - R^T R d_\omega).$$

Practical experiments are repeated a couple of times, with the control gains tuned based on empirical experience, and the control gains of the baseline controller are selected as

$$K_p = \text{diag}([6.0, 6.0, 7.8]), K_v = \text{diag}([8.0, 8.0, 16.0]),$$

$$K_R = 1.92, K_\omega = 0.3.$$

Through computational analysis, one can verify that the selected parameter satisfies the condition formulated in Proposition 1. Furthermore, when implementing the proposed method $u = u_b + u_{L1}$, u_b used the same control gains as the aforementioned baseline controller, their only difference lies in the inclusion or exclusion of $\mathcal{L}_1$ adaptive augmentation. The relevant control gains of the proposed $\mathcal{L}_1$ adaptive controller are selected as

$$A_s = -\text{diag}([65, 65, 65, 80, 80, 80]), T = 0.01, \omega_c = 6.$$

In repeated experiments, the sampling time T is tuned to balance the accuracy of estimation and the proportion of high-frequency components, and the bandwidth ω_c is tuned to balance the smoothness and delay of the $\mathcal{L}_1$ control signal.

Table 2

Quantitative data for experiment 1.

Exp 1	MAX (m)				RMS (m)			
	$ e_x $	$ e_y $	$ e_z $	$\ e\ _2$	e_x	e_y	e_z	$\ e\ _2$
Baseline	0.538	0.141	0.081	0.540	0.247	0.058	0.036	0.256
Proposed	0.323	0.097	0.070	0.332	0.157	0.047	0.032	0.167

Table 3

Quantitative data for experiment 2.

Exp 2	MAX (m)				RMS (m)			
	$ e_x $	$ e_y $	$ e_z $	$\ e\ _2$	e_x	e_y	e_z	$\ e\ _2$
Baseline	0.252	0.139	0.155	0.258	0.134	0.041	0.085	0.164
Proposed	0.184	0.089	0.075	0.187	0.101	0.036	0.038	0.114

5.1. Experiment 1: Manipulator swing test

In this test, a payload with the mass of 50 g is caught by the end effector, the control objective is to stabilize the hex-rotor UAV at the point $p_d = [0, 0, 0.7]^T$ m while the manipulator moves according to curves recorded in Fig. 4a. It is worth mentioning that in all experiments, during both takeoff and hovering, the hexacopter maintained the attitude nearly parallel to the horizon, with the error relative to the desired initial attitude close to 0° , much smaller than 90° , thus satisfying the condition in (25). The results of the two methods are illustrated in Fig. 4, the UAM first hovers at the specified position, and after $t = 2.5$ s (pink area), the manipulator starts to move along the predefined trajectory. It can be observed that both methods can stabilize the system near the desired position, moreover, the proposed method is more effective in mitigating the influence of the manipulator motion. From the uncertainties estimation in Fig. 4d, it can be seen that for the unmodeled end effector payload, the proposed method estimates its moment acting on the y axis and compensates for it. Furthermore, for the estimated uncertainty $\hat{\gamma}_f$ that contains high-frequency components, the control signal becomes smooth after passing through the low-pass filter, and high-frequency components are excluded from the control channel. Table 2 provides the maximum and root mean square (RMS) values of the position errors of the hex-rotor UAV in the test, and one can find that especially along the x axis, which is mainly affected by the swing of the manipulator, the maximum error and RMS error of the proposed method are significantly smaller than those of the baseline method. Therefore, as illustrated in Fig. 4 and Table 2, one can conclude that the proposed controller exhibits better performance in handling the uncertainty induced by the manipulator swing motion and unmodeled payload.

5.2. Experiment 2: Trajectory tracking test

To further verify the performance of the proposed method in trajectory tracking, the UAV is directed to follow a desired trajectory $p_d = [-1.2 \sin(\frac{\pi}{8}t + \frac{5\pi}{6}), -1.2 \cos(\frac{\pi}{8}t + \frac{5\pi}{6}), 0.9]^T$ m, while the manipulator swings periodically as the curves in Fig. 5a. The results of the test are presented in Figs. 5, 6 and Table 3. Based on the results in Figs. 5 and 6, it is obvious that both methods enable the system to follow the desired trajectory, and the proposed method achieves better tracking performance with similar control efforts. The maximum errors and RMS errors by the two methods for the UAV's positions in three directions are summarized in Table 3, from which one can find that the proposed method exhibits smaller tracking errors along all the xyz axes compared to the baseline method. Therefore, the proposed method demonstrates superior trajectory tracking performance.

5.3. Experiment 3: Pick-up and release task

To emphasize the enhancement of manipulation capability for the UAM system, a pick-up and release task is designed. After takeoff, the

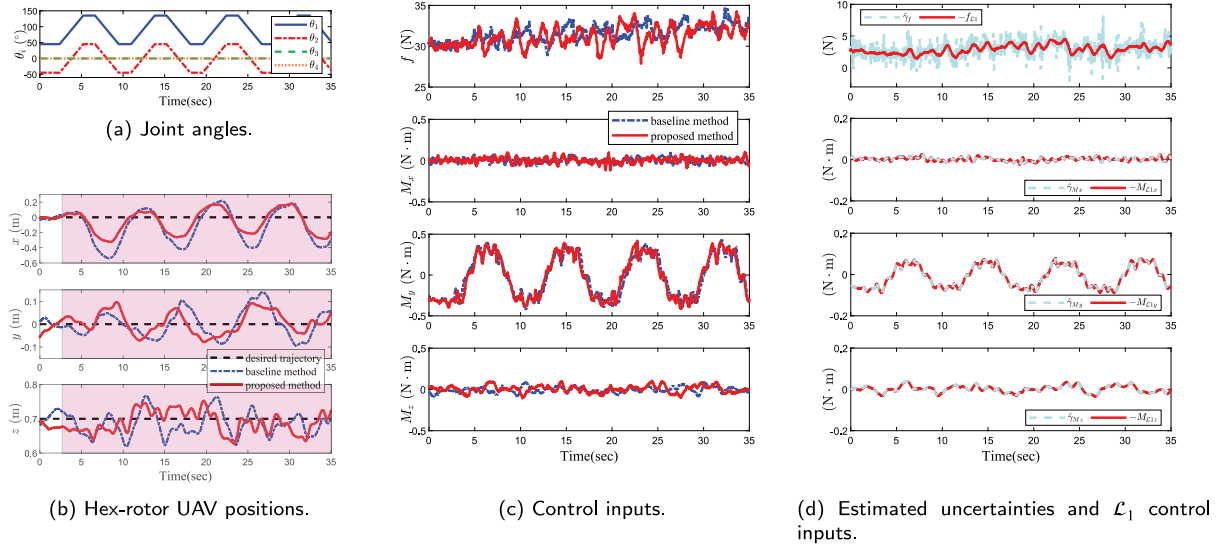


Fig. 4. Results for Experiment 1. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

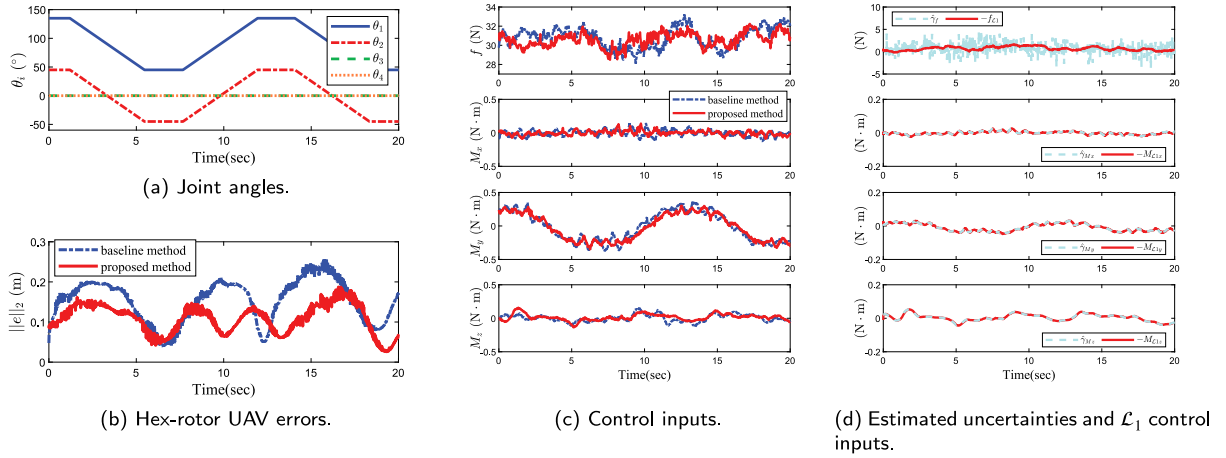


Fig. 5. Results for Experiment 2.

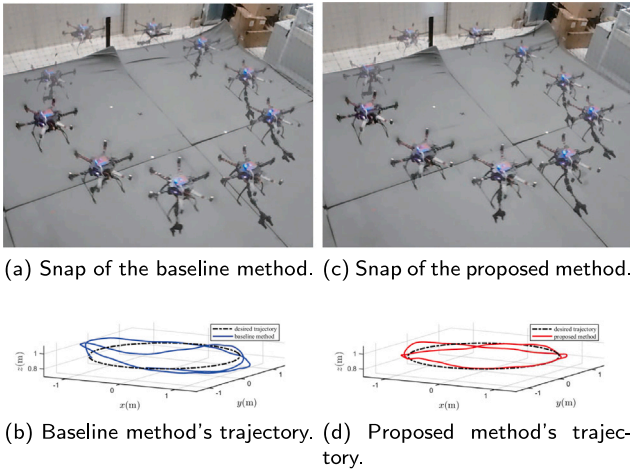


Fig. 6. Trajectories for Experiment 2.

UAM maneuvers to the vicinity of an object and picks it up, and then moves to the desired position to release the object, following the curves

recorded as shown in Fig. 7a, where the pink and green areas are the moments of picking and releasing, respectively. The results depicted in Fig. 7 indicate that the proposed method can rapidly compensate the effects of the increased payload, allowing the UAM to smoothly pick up the object. Several picking experiments are conducted, as illustrated in the accompanying video. Due to the somewhat distance between the object and the UAM's CoM (approximately 45 cm along the x axis), the picking maneuver generates a significant torque about the y axis. For instance, during the picking phase, an unexpected change in the end payload weight leads to instant deviations in the UAM's operation. Consequently, when controlled by the baseline method, the UAM experiences instability due to its inability to effectively compensate the dynamic changes, potentially resulting in collapse. The snap of the pick-up and release process is illustrated in Fig. 8, depicting the successful execution of the task with the proposed method.

6. Conclusions

This paper presents an $\mathcal{L}_1$ adaptive augmentation control method for unmanned aerial manipulator systems, with the goal to overcome uncertainties and dynamic coupling during manipulation tasks. By constructing a detailed dynamic model of the UAM and distinguishing between disturbances that can be directly compensated and those that

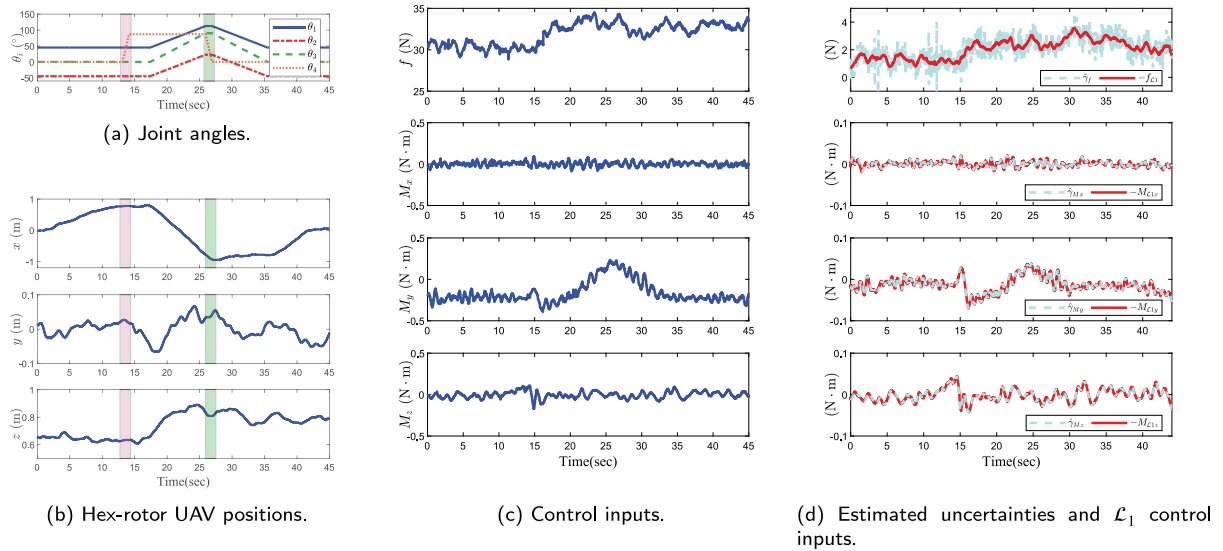


Fig. 7. Results for Experiment 3. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



Fig. 8. The snapshot of the pick-up and release process.

cannot be directly compensated, an integrated control strategy combining geometric control with $\mathcal{L}_1$ adaptive is designed. The theoretical validity of the proposed method is supported by Lyapunov stability analysis, while its practical efficacy is demonstrated through three separate experimental tests on a self-built testbed. The results indicate that the proposed control method offers advantages in enhancing system robustness and improving control precision, thus providing a reliable control strategy for UAM systems to execute operational tasks. In future efforts, impedance control will be considered to facilitate physical interaction operations with the environment while preserving contact.

CRediT authorship contribution statement

Kexin Cai: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Hai Yu:** Writing – review & editing, Methodology. **Zhaopeng Zhang:** Writing – review & editing, Methodology. **Xiao Liang:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Yongchun Fang:** Supervision, Funding acquisition. **Jianda Han:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.conengprac.2025.106418>.

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